

Ultrasonic Range

Time an Echo; publish a qualified reading

TP-E and three LEDs explain the circuit without Serial.

Lesson	019 — component	Time	25–40 minutes
Prerequisites	Lessons 001–018	Board	Arduino Mega 2560
Evidence	Echo/TP-E and D45–D47	Claim	software prepared; specimen gated
Safety	E1: USB-powered low-voltage circuit only		

1 Mission: the range hunt

Make the three result lights answer a quick physical puzzle:

Trigger → Echo width → **RangeState** → visible classification.

First earn the green **Valid** light with a flat card. Then rotate or move the card until the red **Timeout** light wins. If the identified module and available space can produce a complete but excluded pulse, hunt for the amber **OutOfRange** light too. Each later result depends on the same wiring and timing path as the earlier one.

The ranger never turns “no pulse” into zero distance. A complete Echo outside the configured interval is **OutOfRange**; no accepted Echo is **Timeout**; only **Valid** carries a usable distance.

Boundary. This sensor is a learning instrument, not a collision, presence, occupancy, or personnel-safety device. Target shape, angle, material, air conditions, and module variation affect the result.

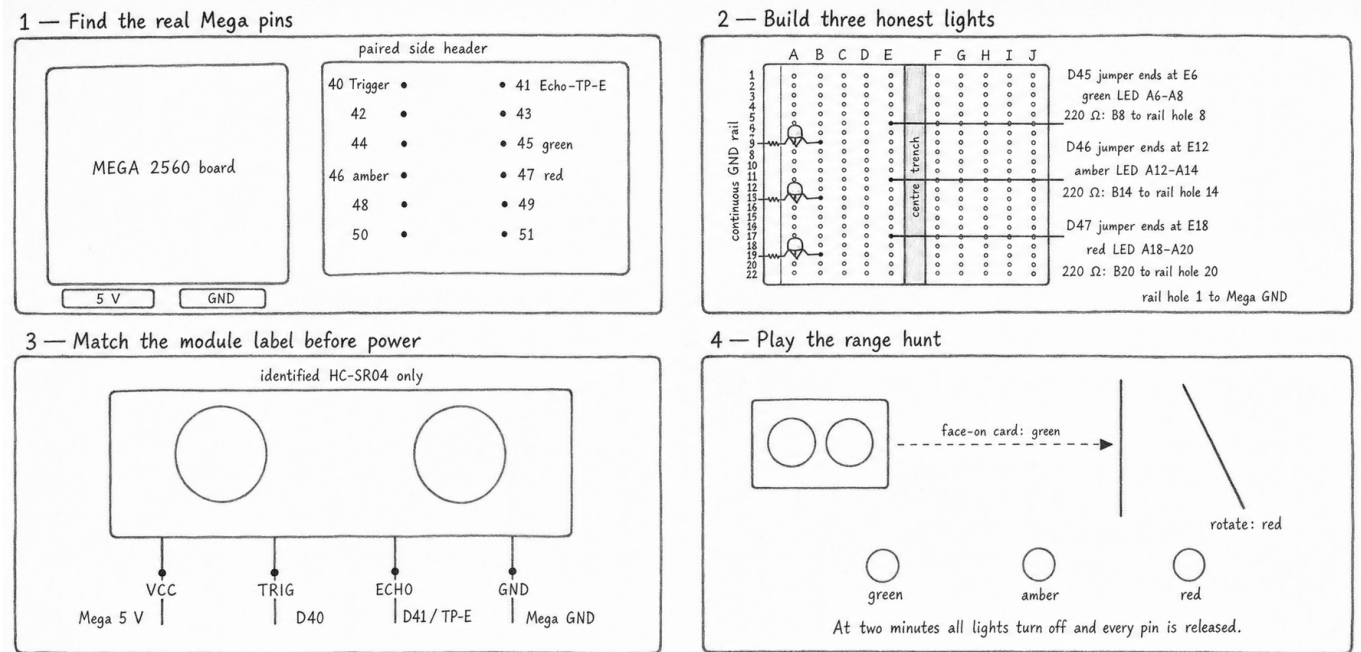
2 Parts, limits, and wiring

Part	Requirement
Mega 2560	USB powered, 5 V logic
Ultrasonic module	Exact identified Elegoo EL-SM-001, or bench work blocked
Evidence LEDs	Three LEDs, each with its own 220 Ω or larger resistor
Target	Flat cardboard, initially perpendicular to the sensor
Instrument	High-impedance logic probe or oscilloscope

Remove USB power before wiring or moving probe clips. Stop for a hot part, unstable USB connection, voltage outside the rails, or disagreement between the module label and this table.

Specimen gate. The repository has no recorded photograph, label, lot, or revision for the learner’s module. The wiring below is therefore a conditional circuit for the Elegoo EL-SM-001, whose manufacturer product page specifies a 5 V DC supply. If the specimen cannot be matched to that identity and its primary documentation, do not power it. Do not infer clone ratings from the HC-SR04-shaped board.

Signal	Mega	Connection
Trigger	D40	module Trigger
Echo / TP-E	D41	module Echo and probe tip
Valid evidence	D45	jumper to a6; LED anode a6, cathode a8; 220 Ω a8 to GND rail
Outside evidence	D46	jumper to a12; LED anode a12, cathode a14; 220 Ω a14 to GND rail
Timeout evidence	D47	jumper to a18; LED anode a18, cathode a20; 220 Ω a20 to GND rail
Supply	5 V, GND	module VCC and GND; probe ground to Mega GND



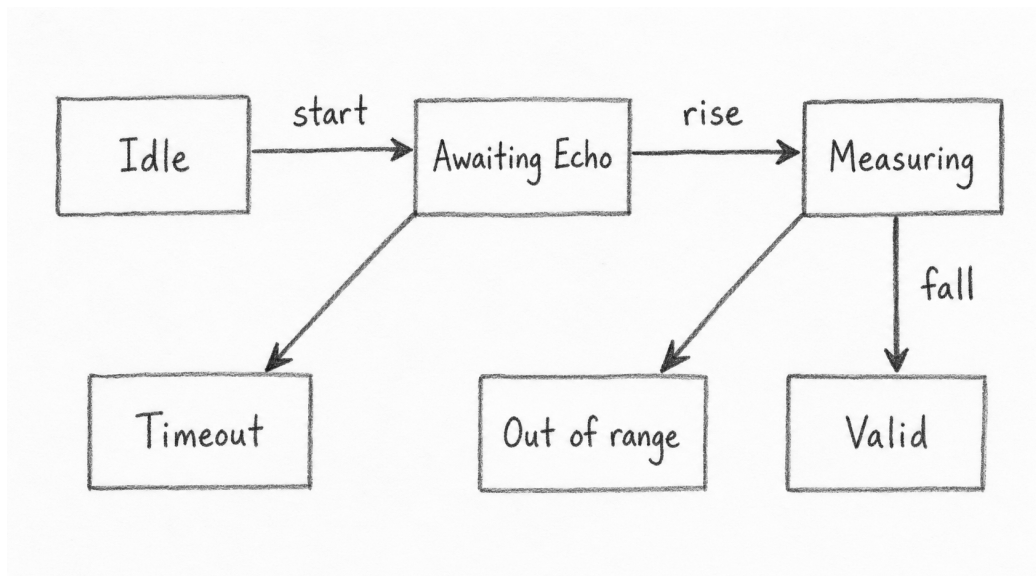
Alternative text: four-stage pencil plate. Stage 1 locates the actual paired Mega digital header and power header. Stage 2 gives exact breadboard rows for three LEDs and individual resistors. Stage 3 connects the identified module's VCC, Trigger, Echo, and GND pins. Stage 4 shows the flat card positions that produce visible Valid, OutOfRange, or Timeout results.

3 Deterministic model boundary

`UltrasonicRanger` is deterministic, non-owning, and hardware independent. The sketch's endpoints own a `DigitalOutput` for Trigger and a `DigitalInput` for Echo. Every update supplies both time and the observed Echo level. Resetting the pure model does not release a pin; endpoint shutdown does.

Operation	Contract
<code>initialize()</code>	return Status ; validate the range and timing configuration
<code>startMeasurement(now, echoHigh)</code>	return Status ; arm from explicit evidence
<code>update(now, echoHigh)</code>	return Status ; advance from supplied evidence
<code>reading()</code>	return stable state, distance, Echo duration, and validity
<code>reset()</code>	retain initialization but restore the inert reading state

Check each returned **Status** with `ok()`; preserve the complete status when reporting failure. The configured values are 30 ms Echo timeout, 30 ms maximum Echo duration, 20–4000 mm accepted distance, and 343 $\mu\text{m}/\mu\text{s}$ assumed sound speed. These are software classification bounds and a model assumption. They do not certify accuracy, prove a specimen identity, or restate the manufacturer's full operating envelope.



Pencil state map: Idle starts Awaiting Echo. An Echo rise starts Measuring; its fall produces Valid. Awaiting Echo can instead produce Timeout, and Measuring can instead produce Out of range.

4 Narrative run loop

```

const adk::MicrosecondTimePoint now (micros ());

if (!observeRange (now))
{
    stopSafely ();
    return;
}

const RangeSignal signal = decideRangeSignal ();

if (!showRangeSignal (signal))
{
    stopSafely ();
    return;
}
  
```

Objects appear as runtime, endpoints, evidence LEDs, then ranger. Setup acquires them in that order. The three LEDs illuminate together for 150 ms: an immediate “the adventure started” sparkle that also exposes a dead or backwards LED before ranging begins. The loop then observes, decides, actuates, and requests a new reading no sooner than every 60 ms. Idle, valid, timeout, and out-of-range are all eligible for that next request; a valid distance cannot stop the measurement cadence and become silently stale.

The 10 μ s Trigger pulse uses one explicit bounded delay. Echo timing and the measurement cadence remain nonblocking. After two minutes, an input-independent deadline drives Trigger low and releases every endpoint even if Echo is stuck or absent. No library behavior allocates, throws, or reads wall-clock time.

5 Four visible wins

5.1 Win 1: all three lights sparkle

Upload, then watch D45–D47 illuminate together for 150 ms. If one is missing, fix its polarity, row, or resistor before moving on. The sparkle proves only that the visible path started; the next win must come from a real Echo.

5.2 Win 2: make green follow the card

Attach probes while unpowered. For a specimen whose primary documentation calls for it, place the flat card about 200 mm away and perpendicular to the transducers. Green D45 should follow only after a 10 μ s Trigger pulse on D40 and a later high interval on D41/TP-E. Move the card to roughly 500 mm and 1000 mm; the light should stay green while Echo becomes wider.

Target	Trigger width	Echo width	LED result
200 mm	_____	_____	_____
500 mm	_____	_____	_____
1000 mm	_____	_____	_____

Estimate one distance using

$$d_{\text{mm}} \approx \frac{t_{\text{echo}, \mu\text{s}} \times 343}{2000}.$$

Division by two accounts for the outward and return path. A plausible estimate unlocks the reason behind the light; it is not calibration.

5.3 Win 3: make red win without rewiring

Turn the flat target away without touching the powered circuit. A missing accepted Echo should select red D47. Return the card face-on and green should recover by itself. That recovery is the proof: the same wiring distinguishes “no accepted Echo” from a usable measurement.

5.4 Win 4: hunt amber, then watch the finale

Only when space and the identified specimen’s documentation permit it, seek a complete Echo whose calculated distance falls outside 20–4000 mm. That selects amber D46; failure to find one is not a reason to invent evidence or exceed the module’s documented setup.

At two minutes all three LEDs turn off automatically. D40 is driven low before it becomes input/high impedance, TP-E is no longer sampled, and every claim is released. Check Trigger electrically if a probe is already attached. An LED being off does not by itself prove the pin’s safe state.

6 Diagnosis

Observation	Check next
No startup light walk	endpoint conflict, polarity, resistor, and initialization status
Trigger present; Echo absent	module supply, common ground, target angle, and exact module
Echo present; fault LED	pulse width, timeout boundary, and probe reference
Valid LED; implausible distance	target geometry, Echo width, sound-speed assumption, units
Outside LED at ordinary range	configured limits and integer conversion
Trigger remains driven after stop	safe-state failure; remove USB and record the deviation

7 Keep exploring

The HTML reference carries searchable API notes, corrections, and the canonical sketch. The learner adventure remains the visible sparkle, green/red recovery, optional amber hunt, and automatic finale.

- UltrasonicRanger public header and implementation
- PulseInput public header and implementation
- canonical Mega 2560 sketch

Publication boundary. No physical observation is claimed; the exact specimen gate remains in force.

8 Primary references

- Elegoo EL-SM-001 product specification: conditional source for the named module’s 5 V supply, stated range, acoustic frequency, and model identity.
- Elegoo HC-SR04 documentation index: manufacturer-hosted datasheet route; match it to the physical specimen before bench work.
- Arduino Mega 2560 Rev3 datasheet: official board power and D40–D47 connector identities.

9 More range games

1. Make a hand-drawn map of where green survives as the card rotates.
2. Compare a book, a soft cloth, and corrugated cardboard without changing wiring or code.
3. Race to predict which light appears before each target move.
4. Explain why turning every invalid result into “zero” would ruin the game.